Discussion Exercise 6

University of California, Riverside
July 13, 2023

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1. Torque

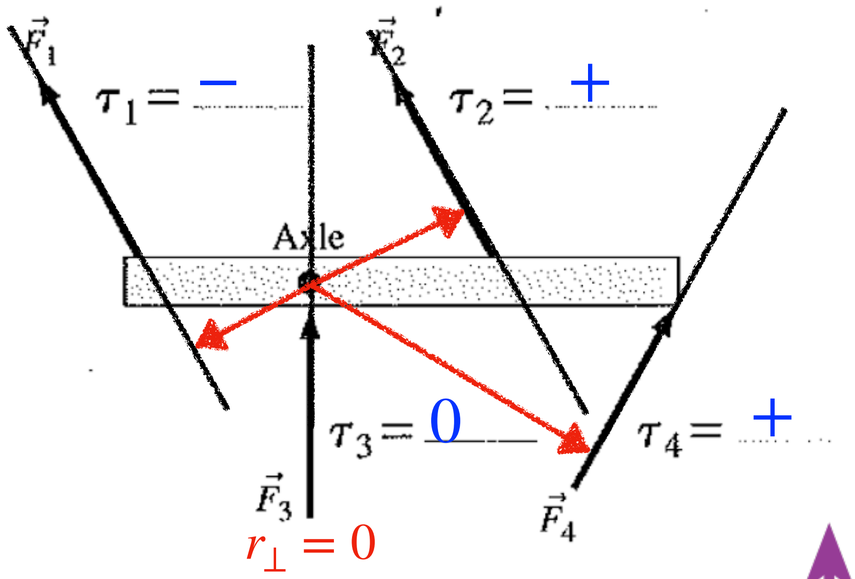
Four forces are applied to a rod that can pivot on an axle. For each force,

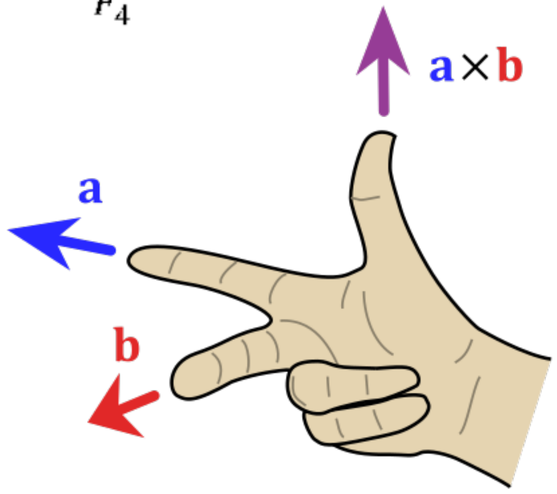
a. Use a **black** pen or pencil to draw the line of action.

b. Use a **red** pen or pencil to draw and label the moment arm, or state that $r_\perp = 0$.

c. Determine if the torque about the axle is positive (+), negative (−), or zero (0).

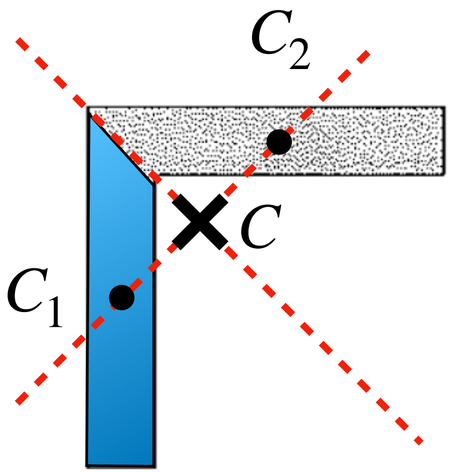
Choose the positive direction to be out of the plane, then use the definition $\vec{\tau} = \vec{r} \times \vec{F}$





2. Center of Gravity

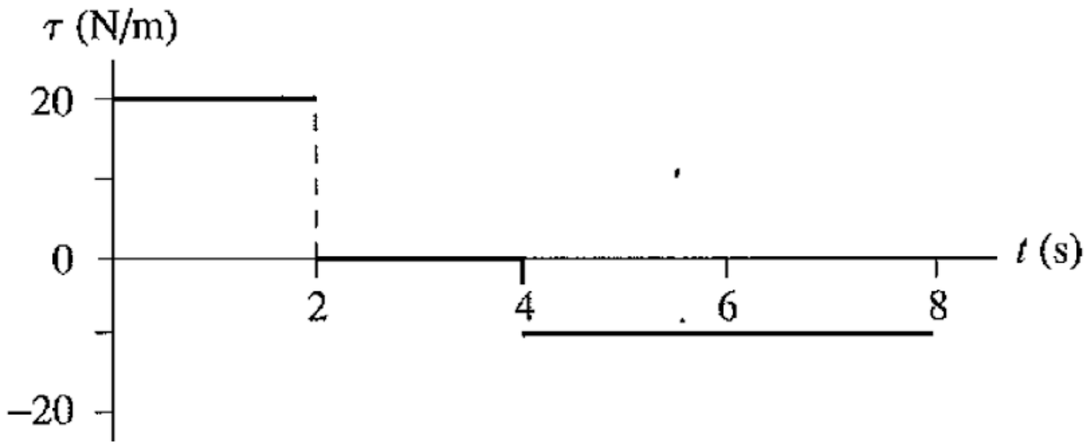
Mark the center of gravity of this object with an ×. Explain.



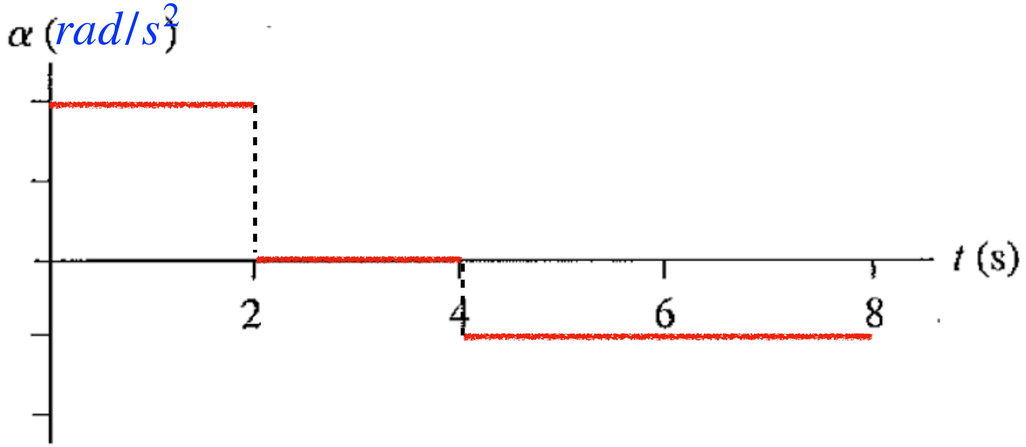
Notice that this object is symmetric wrt the diagonal dashed line. So we can first determine the center of mass C_1 and C_2 for each individual piece. Then the center of mass of the entire system is just intersection point of the line connecting C_1 and C_2 with the diagonal dashed line

3. Rotational Dynamics

The top graph shows the torque on a rotating wheel as a function of time. The wheel's moment of inertia is $10\ \mathrm{kg \cdot m^2}$. Draw graphs of α-versus-t and ω-versus-t, assuming $\omega_0 = 0$.



Based on New's second law for rotation $\vec{\tau} = I\vec{\alpha}$, the angular acceleration is given by $\alpha = \tau/I$. We can use this formula to draw the $\alpha - t$ graph.



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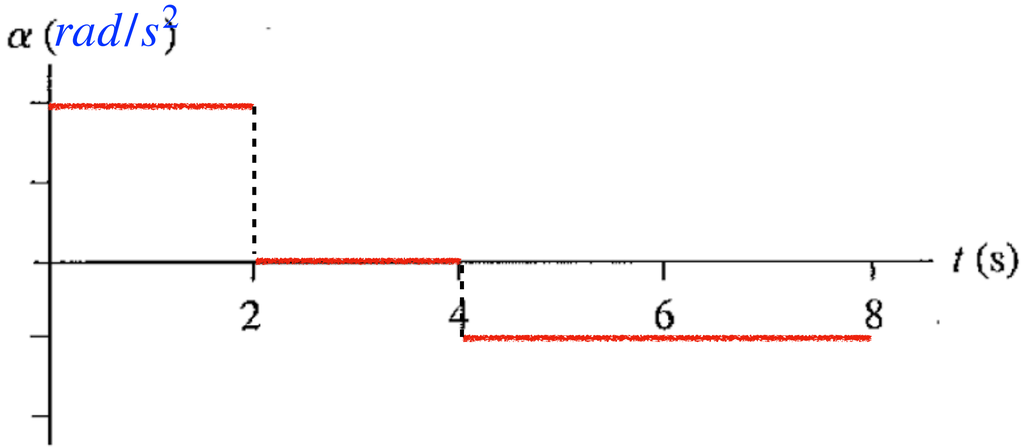
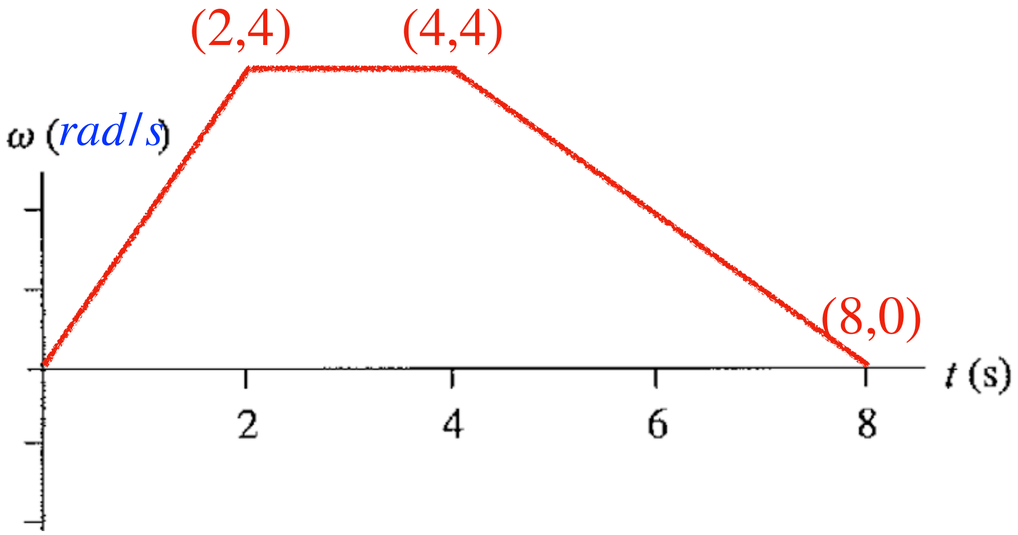
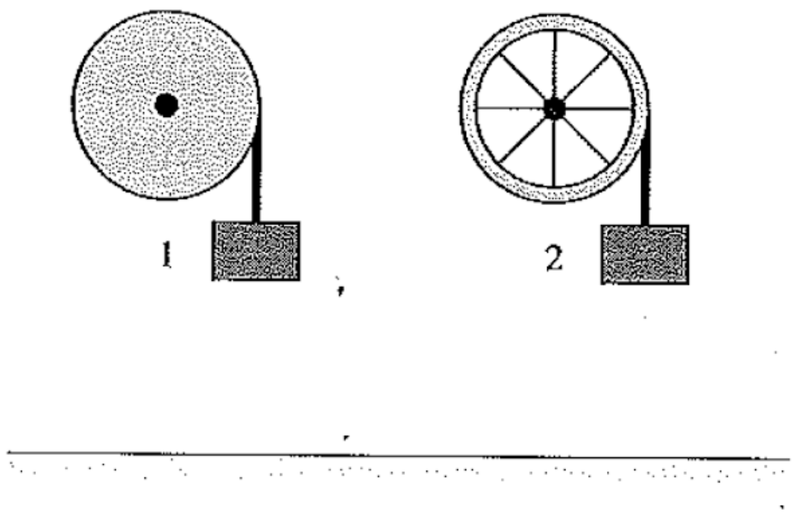




Notice that $\alpha = d\omega/dt$, the integral of angular acceleration wrt time is $\Delta\omega = \int_{t_i}^{t_f} \alpha dt$, which means the the change in the angular frequency is just the area under the $\alpha - t$ curve!

4.Newton's Law of Gravity

A solid cylinder and a cylindrical shell have the same mass, same radius, and turn on frictionless, horizontal axles. (The cylindrical shell has lightweight spokes connecting the shell to the axle.) A rope is wrapped around each cylinder and tied to a block. The blocks have the same mass and are held the same height above the ground. Both blocks are released simultaneously. The ropes do not slip.



Which block hits the ground first? Or is it a tie? Explain.

The left one will the the ground first.

1. To determine which block hits the ground first, we need to compare their angular acceleration: the one with a bigger angular acceleration will hit the ground first.

2. Notice that the torques are the same wrt their own axles. Then according to Newton's law of rotation $\tau = I\alpha$, α is inversely proportional to the moment of inertia if their torques are the same. Consequently, the relation $I_1 < I_2$ immediately leads to $\alpha_1 > \alpha_2$. Note: here the momentum of inertia I_{1or2} should be that of the entire system including the block.